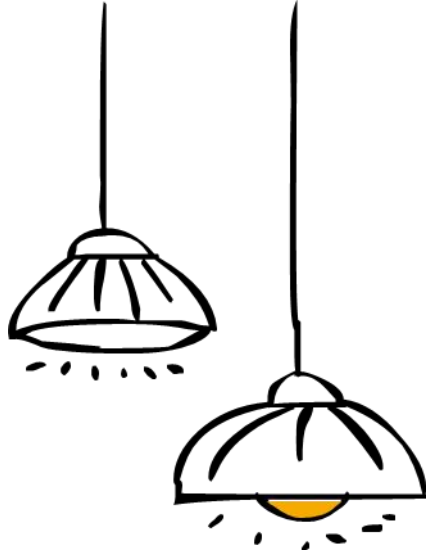
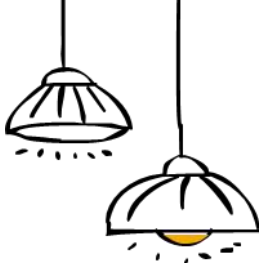




# Data Types in Data Analytics

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# Lecture Outline

- Explore the importance of data type
- Compare different types of business data
- Explain the data analysis tips
- Discuss business data case studies



# Why Data Types Matter?

- **Data's DNA:** Before analyzing data, a manager must understand its "DNA."
- **Proper Analysis:** Data types dictate which statistical tests are valid and which visualizations are appropriate.
- **Garbage In, Garbage Out:** Incorrectly identifying a data type leads to flawed models.
- **Storage & Cost:** Different types require different storage architectures (SQL vs. NoSQL).



# Quantitative (Numerical) Data

- Quantitative data represents measurable quantities.
- It is further divided based on the "fineness" of the measurement.
- **A. Discrete Data**
- **B. Continuous Data**



# Discrete Data: the countable data

- Definition: Countable values with no "in-between" (integers).
- Discrete data represents values you can count one by one.
- No decimals or fractions are allowed.
- Usually used for counting people, objects, or events.
- Example: the number of students in a class (30, 31, 32) or how many books you have (5, 6, 7)
- Business Example: Number of customer complaints, daily store visitors, or inventory units.



## Discrete Data – Analytic Tips

- Visualized with Bar Charts because values are separate and countable.
- Frequency tables work well for summarizing how frequent each value occurs.
- Often analyzed using counts, proportions, or percentages.
- Good for comparing categories or distinct groups.
- Example analyses: number of customers per day, number of items sold in each category.



# Continuous Data: Measurement

- Continuous data represents values you can measure.
- These values can take any value within a range, including decimals.
- Can be infinitely precise depending on the measuring tool.
- Usually used for measuring time, amount, distance, or size.
- Example: someone's height (150.5 cm), or the temperature (29.8°C)
- Business Examples: sales revenue amount, delivery time in hours, product weight, machine operating time, etc.



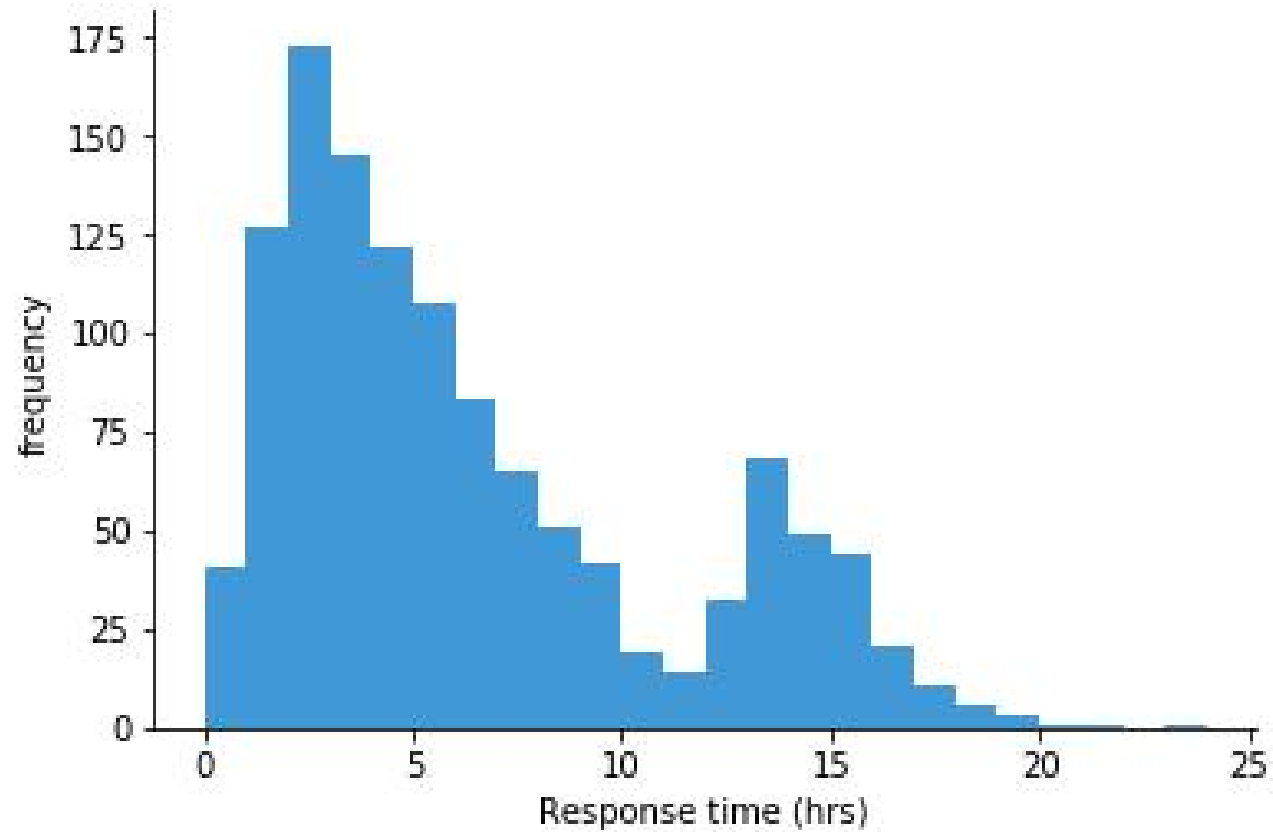
## Continuous Data – Analytic Tips

- Visualized with Histograms or Line Charts - the data flows across a range.
- Descriptive statistics (mean, median, standard deviation) are applied.
- Useful for understanding trends, patterns, and distributions.
- Often used in forecasting and modeling - the values change smoothly.
- Example analyses: sales revenue over time, temperature trends, processing time in minutes.





# Histogram for Continious data





# Qualitative (Categorical) Data

- Qualitative data refers to information that describes qualities, characteristics, or categories rather than numbers.
- They help us understand what type or which group something belongs to.
- They are not measured with numbers; instead, they represent labels or categories used for classification in business.
- **A. Nominal Data**
- **B. Ordinal Data**



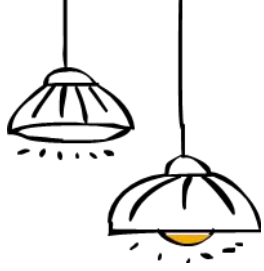
# Nominal Data (Labels Without Order)

- Nominal data represents names or labels with no ranking or order.
- Categories are different, but none is “higher” or “lower.”
- Common in business for identifying customer groups, product categories, or market segments.
- Examples: customer gender, product brand, payment method, industry type.



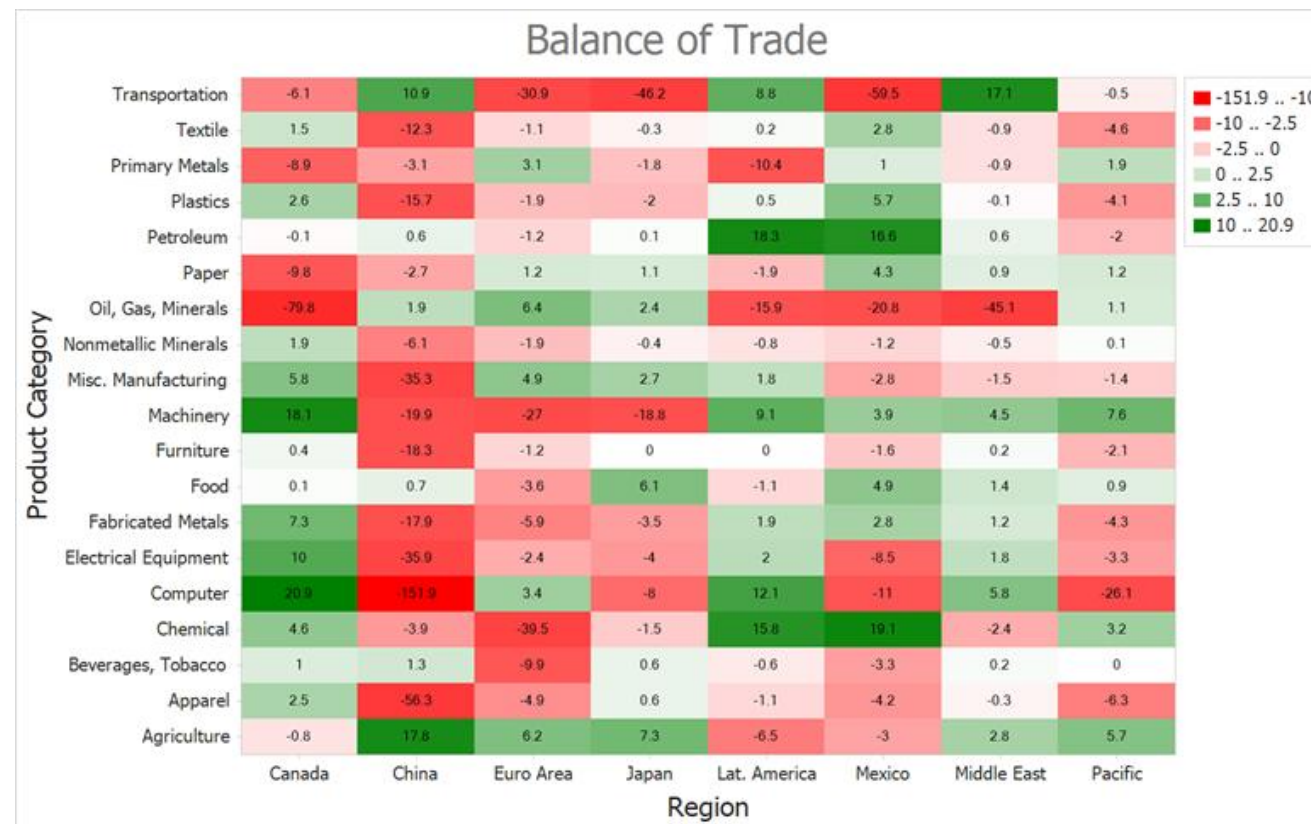
## Nominal Data - Analytic Tips

- Usually visualized with Bar Charts or Pie Charts or Heatmaps because each category is separate.
- Best summarized using frequencies and percentages.
- Works well for customer segmentation, brand comparison, and preference analysis.



# Heatmaps in Nominal data

- Example: Usage: Cross-tabulating two categorical variables (e.g., Region vs. Product Line)



Source: raw.githubusercontent.com



# Ordinal Data: Ordered Category Data

- Ordinal data uses categories, but these categories have a meaningful order.
- The order matters (e.g., “High” is above “Medium”), but the gaps between levels are not equal.
- Common in business for ratings, satisfactions, rankings, and priority levels.
- Examples: satisfaction levels (Poor → Fair → Good → Excellent), education level, service priority.

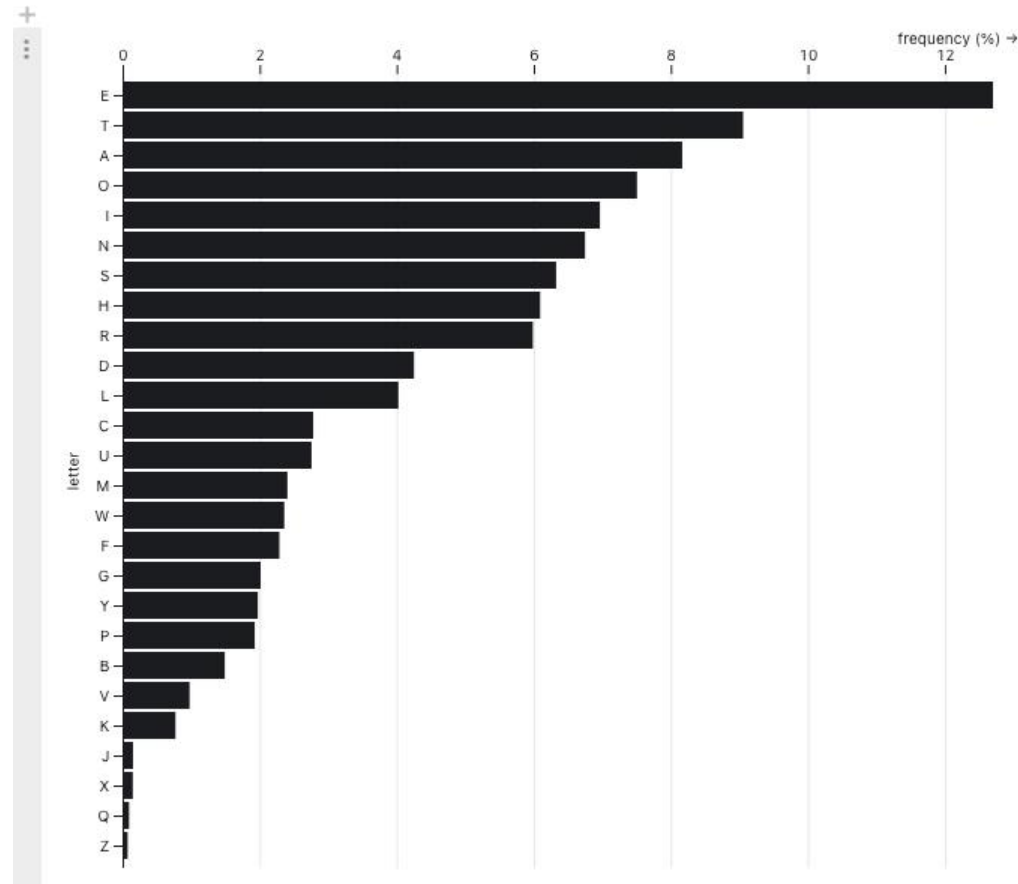


## Ordinal Data - Analytic Tips

- Usually visualized with Ordered Bar Charts to show ranking clearly.
- Best described using the median or mode, not the mean.
- Useful for analyzing customer satisfaction, service quality, and preference intensity.

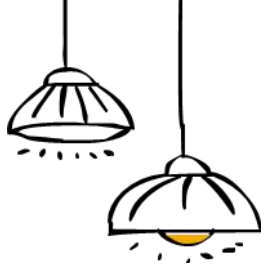


# Ordered Bar Charts for Ordinal data



Source: observablehq.com





# Advanced Classification: The 4 Levels of Measurement

- To perform advanced analytics, we use Stevens' Levels of Measurement.
- Classify data into four types based on how much mathematical meaning each type carries.

**A. Nominal**

**C. Interval**

**B. Ordinal**

**D. Ratio**

- As we move from Nominal → Ordinal → Interval → Ratio, the data become more powerful for statistical analysis.



# Nominal Level (Category Labels Only)

- Represents names, labels, or groups with no order.
- Only shows difference, not ranking.
- Best summarized using counts and percentages.
- No arithmetic possible.



## Ordinal Level (Ordered Categories)

- Represents categories that have a meaningful order, but differences between levels are not measurable.
- Analyze with median or mode.
- Cannot calculate mean meaningfully.



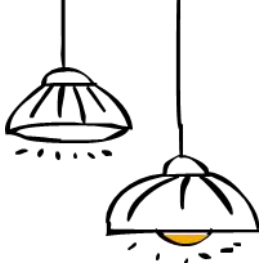
# Interval Level (Equal Intervals, No True Zero)

- Numeric data where the difference between values is meaningful, but there is no true zero point.
- Zero does not mean “nothing.”
- Examples: temperature in Celsius or Fahrenheit, dates, calendar years.
- Use Line Charts and Histograms.
- Can calculate mean and standard deviation.

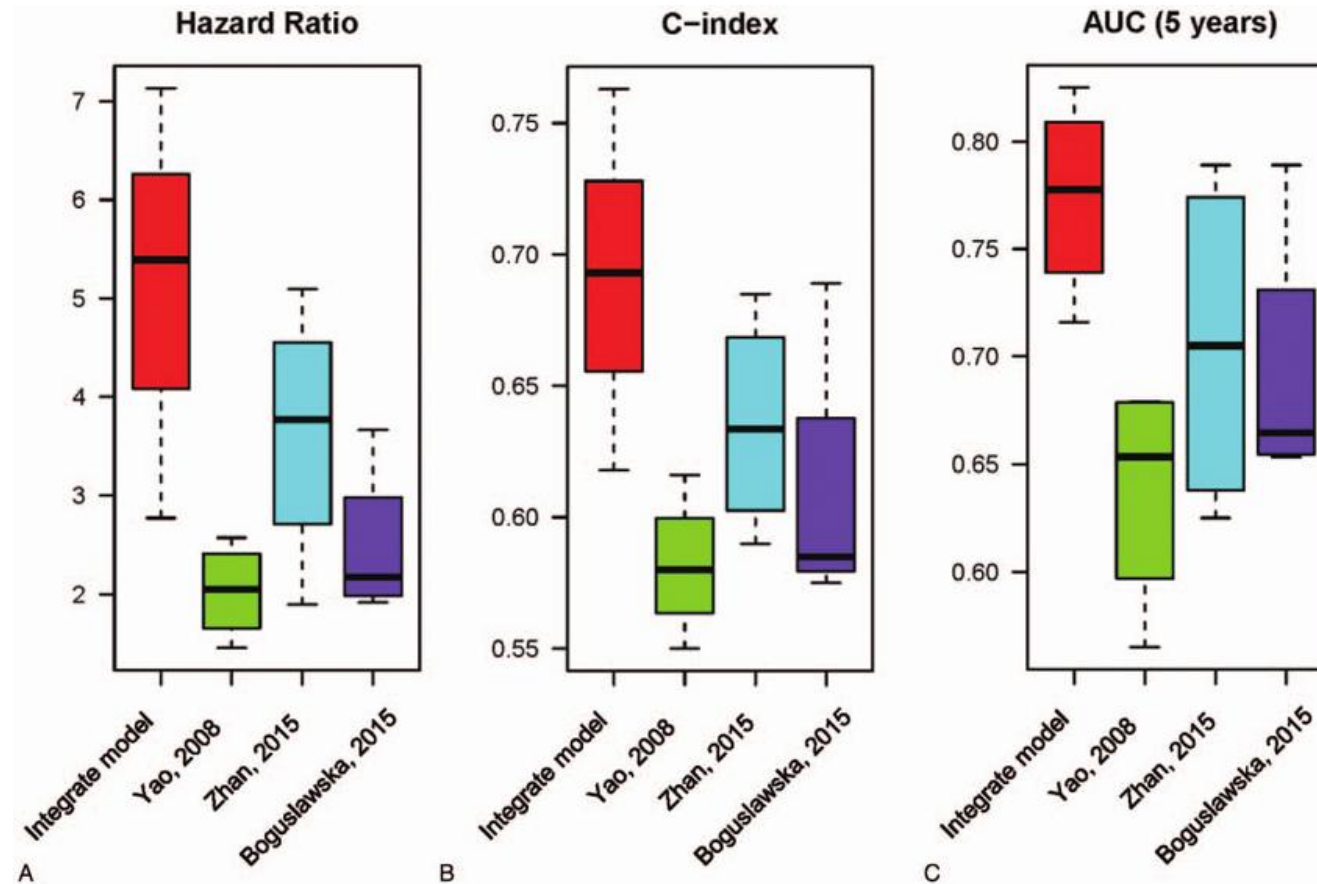


# Ratio Level (Highest Level – True Zero Exists)

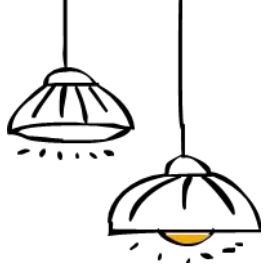
- Numeric data with equal intervals and a true zero, meaning zero represents “none.”
- Ratios are meaningful (20 kg is truly twice 10 kg).
- Examples: revenue, weight, height, age, distance, number of sales.
- Visualized with Histograms, Line Charts, Box Plots.
- Supports all statistical techniques: mean, variance, correlations, regression.



# Box plot for Ratio data

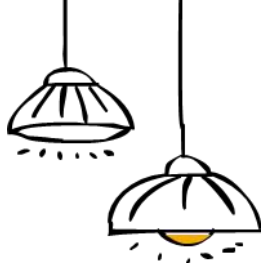


Source: [https://www.researchgate.net/figure/Box-plot-of-3-indicators-hazard-ratio-HR-C-index-AUC-in-different-datasets-A-Box\\_fig2\\_328675442](https://www.researchgate.net/figure/Box-plot-of-3-indicators-hazard-ratio-HR-C-index-AUC-in-different-datasets-A-Box_fig2_328675442)



# Comparison

| Level    | Characteristics          | Operations               | Business Example     |
|----------|--------------------------|--------------------------|----------------------|
| Nominal  | Labels only              | Counting, Mode           | Product Categories   |
| Ordinal  | Order matters            | Median, Percentiles      | Star Ratings (1-5)   |
| Interval | Order + Equal gaps       | Addition/ Subtraction    | Temperature, Dates   |
| Ratio    | Order + Gaps + True Zero | Multiplication/ Division | Revenue, Salary, ROI |



# The Visualization Selection Matrix

| Target Analytics | Data Type                  | Recommended Chart          |
|------------------|----------------------------|----------------------------|
| Comparison       | Nominal / Ordinal          | Bar Chart, Clustered Bar   |
| Composition      | Nominal (Parts of a whole) | Stacked Bar, Donut Chart   |
| Distribution     | Continuous                 | Histogram, Box Plot        |
| Relationship     | Ratio / Interval           | Scatter Plot, Bubble Chart |
| Trend            | Time + Ratio               | Line Chart                 |





# Unusual Data Structure

- In business and data analytics, data comes in different forms and formats, and understanding these types is crucial for making effective decisions.
- Broadly, data can be quantitative or qualitative, and for practical analysis, it is often classified as unusual structures.
- **Structured, Semi-structured, and Unstructured**
- This classification helps us determine how the data can be stored, accessed, and analyzed.



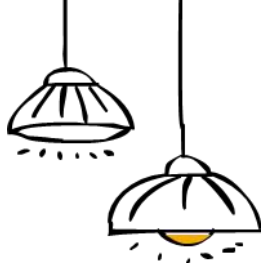
# Structured Data

- Structured data is information that is organized, searchable, and stored in fixed formats such as tables, rows, and columns.
- It fits neatly into databases and spreadsheets, making it easy for analysts to query and run calculations.
- In business, this includes data like sales numbers, customer IDs, transaction records, inventory lists, and performance metrics.

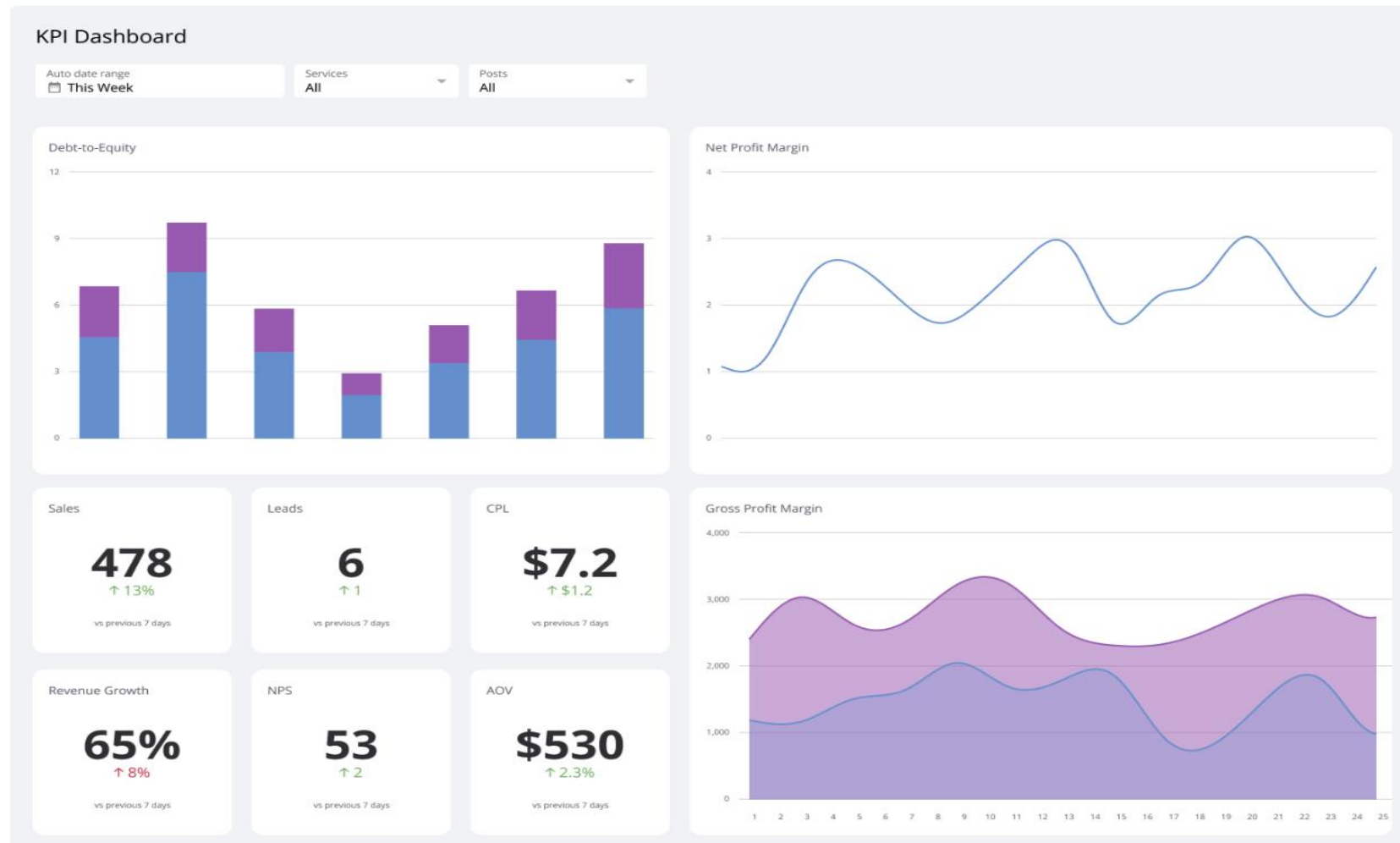


# Structured Data - Analytic Tips

- Best analyzed using SQL queries, dashboards, and statistical models.
- Works well with BI tools because it is clean, consistent, and machine-readable.
- Easy to visualize using tables, bar charts, line charts, and KPIs dashboard.



# KPI dashboard



Source: images.klipfolio.com



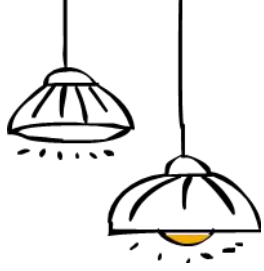
# Unstructured Data

- Unstructured data has no fixed format, making it harder to store, search, and analyze directly.
- This includes text, images, audio, video, emails, customer reviews, chat messages, and social media posts.
- Businesses use unstructured data to understand customer behavior, sentiment, and emerging patterns that are not captured in numeric tables.

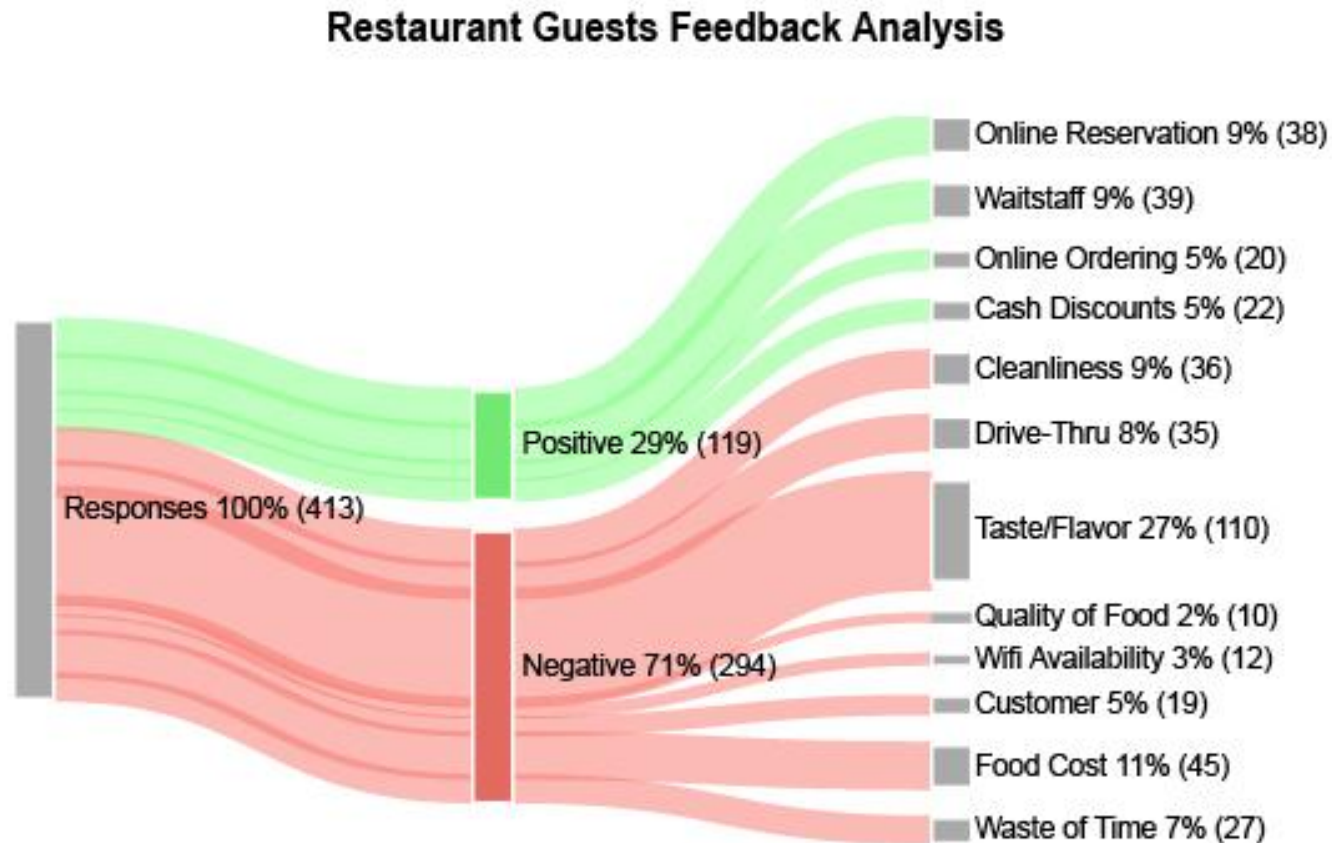


# Unstructured Data - Analytic Tips

- Requires advanced analytics such as natural language processing (NLP), text mining, image recognition, or AI-based tools.
- Useful for discovering customer emotions, trends, opinions, and hidden insights.
- Best visualized through word clouds, sentiment charts, tag clusters, or dashboards combining text and numbers.



# Sentiment Charts in Unstructure data

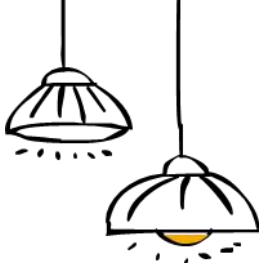




# Semi-Structured Data

- Partially organized; has some structure but not fixed like a database table.
- Examples: XML files, JSON data, emails with headers, online forms.
- Analytic Tip: Can be parsed and analyzed with Python programming or BI tools.
- Visualize using tables, charts, or dashboards after extraction.





# Comparison of Unusual data

| Feature             | Structured Data            | Semi-Structured Data       | Unstructured Data                            |
|---------------------|----------------------------|----------------------------|----------------------------------------------|
| Format              | Fixed rows & columns       | Partially organized        | No fixed format                              |
| Storage             | Databases, spreadsheets    | XML, JSON, emails          | Text, images, audio, video                   |
| Ease of Analysis    | Easy                       | Moderate                   | Hard                                         |
| Examples            | Sales, inventory, IDs      | Email headers, XML, JSON   | Social media posts, videos                   |
| Visualization       | Tables, charts, dashboards | Tables, charts, dashboards | Word clouds, sentiment charts, AI dashboards |
| Tools for Analytics | SQL, BI tools, Excel       | Parsing scripts, BI tools  | NLP, AI, Machine Learning                    |



# Discussion Questions

- The Question: An HR Director wants to track "Level of Education" (High School, Bachelors, Masters, PhD) and "Current Salary." What is the most accurate classification for these?



# Discussion Questions

- Question: An HR Director wants to track "Level of Education" (High School, Bachelors, Masters, PhD) and "Current Salary." What is the most accurate classification for these?
- **Answer:**
- Education has a logical rank (Ordinal), and
- Salary is a measurable amount with a true zero (Ratio).



## Discussion Questions

- Context: You are a consultant for a global retailer. You are looking at a spreadsheet containing "Shipping Method" (Express, Standard, Economy) and "Shipping Cost" (\$).
- The Question: If the CEO asks for a report on which shipping method is the most cost-efficient? Please types of data are there?



# Discussion Questions

- Context: You are a consultant for a global retailer. You are looking at a spreadsheet containing "Shipping Method" (Express, Standard, Economy) and "Shipping Cost" (\$).
- Question: If the CEO asks for a report on which shipping method is the most cost-efficient, how do you handle these two data types?
- **Answer:**
- Shipping Method is Ordinal (there is a clear hierarchy of speed/priority).
- Shipping Cost is Continuous.



## Discussion Questions

- Context: Your HR department wants to predict which employees are likely to quit. They give you a dataset with: Employee Tenure (Years), Department, and Last Performance Rating (1–5).
- The **Question**: Which of these variables can you perform math on (like calculating an average), and which can you only use for grouping?



# Discussion Questions

- **Context:** Your HR department wants to predict which employees are likely to quit. They give you a dataset with: Department, and Last Performance Rating (1–5).
- **Answer:**
- Department is Nominal; you can only use it to group/filter (e.g., "Is turnover higher in Sales?").
- Performance Rating is Ordinal.



## Discussion Questions

- Question: Your HR database assigns every employee a unique ID number (e.g., #102, #103). If you calculate the average of these ID numbers, is the result meaningful? Why or why not?





## Discussion Questions

- Question: Your HR database assigns every employee a unique ID number (e.g., #102, #103). If you calculate the average of these ID numbers, is the result meaningful? Why or why not?
- Answer: No. Even though they are digits, Employee IDs are Nominal. They are labels, not measurements. Averaging them would be like averaging social security numbers; it provides no insight into the "average employee."



## Discussion Questions

- Question: A marketing team collects customer reviews and comments from social media about a new product. What type of data is this?



# Discussion Questions

- Question: A marketing team collects customer reviews and comments from social media about a new product. What type of data is this?
- **Answer:**
- This is Qualitative - Unstructured Data because the information is text-based, descriptive, and does not have a fixed format.



# THANK YOU

Please email me if you have any queries